

**Grading Guide**  
**Programming for Economists**  
**Winter Exam 2025/26**

# Data Analysis Project

## Problem 1: Aggregate Inflation in Denmark

### Question 1.1

- Retrieves PRIS113 data via an API call.
- Constructs the CPI index (2020 = 100).
- Computes month-to-month and 12-month inflation rates.
- Produces clear time series plots with labeled axes and legend.
- Provides a short interpretation of when the inflation surge ended.

### Question 1.2

- Correctly defines and computes the weights  $\kappa(k, \alpha)$  for  $\alpha = 0, 1, 2, 3$ .
- Computes instantaneous inflation using a rolling window or equivalent method.
- Produces clean comparative plots for each  $\alpha$ .
- Discusses how the timing of the inflation surge differs from Question 1.1.

### Question 1.3

- Retrieves PRIS111 data for total CPI, CPI excl. energy, and core inflation.
- Correctly computes 12-month inflation rates.
- Plots the three series clearly since 2019.
- Briefly discusses likely drivers of the post-pandemic inflation surge.

### Question 1.4

- Retrieves all 4-digit PRIS111 product categories.
- Computes 12-month inflation for each category with `groupby` methods.
- Produces percentile plots (25th, 50th, 75th).
- Produces a histogram of price changes from Aug. 2020 to Aug. 2025.
- Identifies and illustrates the top-10 and bottom-10 categories.
- Gives a short descriptive interpretation of the distributional patterns.

## **Problem 2: International Comparison**

### **Question 2.1**

- Retrieves CPI from Denmark Statistics and HICP from FRED.
- Plots the two indexes on a comparable scale.
- Comments on similarities, differences, and whether the indices are comparable.

### **Question 2.2**

- Retrieves HICP data for Denmark, Austria, Euro Area, and the U.S.
- Plots index levels and 12-month inflation rates since 2019.
- Computes min, max, and mean inflation rates year-by-year.
- Summarises cross-country differences in inflation paths clearly.

## **Problem 3: Extension**

- Extends the analysis with at least one new dataset,
- Produces 1–3 new figures with clear economic relevance,
- Explains the insight yielded by the extension.

# Model Analysis Project

## Problem 1: Labor Supply

### Question 1.1

- Evaluates and plots the utility function over a grid for  $\ell_i$ .
- Evaluates and plots the FOC  $\varphi(p_i, c_i, \ell_i)$  over the same grid.
- Finds optimal labor supply using a numerical optimizer.
- Finds optimal labor supply using a root finder.
- Compares the two methods and comments on speed and robustness.
- Compiles results into one clear figure (with subplots allowed).
- Repeats the exercise for the additional  $\epsilon$  values with minimal new code.

### Question 1.2

- Computes the optimal labor supply function  $\ell^*(p_i)$  for  $p_i \in [0.5, 3.0]$ .
- Correctly handles the minimum feasible hours condition.
- Produces clear plots for both  $\zeta = 0.1$  and  $\zeta = -0.1$ .
- Briefly comments on how the lump-sum component affects labor supply.

## Problem 2: Public Good

### Question 2.1

- Draws a productivity distribution and computes labor supply for all workers.
- Computes tax revenue  $T(\tau, \zeta)$  and the Social Welfare Function.
- Repeats this across the required range of  $\zeta$  values and  $\tau$ .
- Plots revenue and SWF clearly.
- Indicates the approximate maximizing point(s).

### Question 2.2

- Runs a numerical optimizer to find  $(\tau^*, \zeta^*)$ .
- Checks robustness to starting values.
- Produces local plots of  $T(\tau, \zeta)$  and SWF around the optimum.
- Computes and plots the Lorenz curve of consumption at the optimum.

### Problem 3: Top Tax

- Evaluates and plots the utility function under the top-tax system.
- Computes and plots the FOC for the pre-kink and post-kink regions.
- Finds optimal labor supply via numerical optimization.
- Computes optimal labor supply via the four-step FOC approach.
- Compares the two methods and comments on speed and differences.
- Repeats the analysis for the additional productivity values with minimal new code.
- Presents results clearly, optionally in a multi-panel figure.

### Question 3.2

- Computes  $\ell^*(p_i)$  and  $c_i$  for  $p_i \in [0.5, 3.0]$ .
- Classifies workers into  $\ell_b$ ,  $\ell_k$ , and  $\ell_a$  cases.
- Reports the proportions in each region.
- Plots both labor supply and consumption clearly.

### Question 3.3

- Computes the SWF under the introduced top tax.
- Computes and plots the Lorenz curve of consumption.
- Explores alternative  $(\omega, \kappa)$  values and reports if SWF improves.

### Problem 4: Extension

- Implements a meaningful model extension (preferences, heterogeneity, tax rules, etc.).
- Evaluates the extended model correctly.
- Explains the resulting economic insight clearly

# Exam Project

## Problem 1: Danish House Prices

### Question 1.1

- Retrieves EJ56 data via an API call and drops provinces with missing values in the sample.
- Correctly constructs a house price index per province.
- Produces a clear time series plot of indexed prices by province and reports the ranking by total growth.

### Question 1.2

- Retrieves CPI data from PRIS113 via e.g. an API call and aligns it with the house price sample.
- Correctly constructs real house prices.
- Produces a clear plot of real house prices by province.

### Question 1.3

- Reads BM010\_houses.xlsx, keeps municipalities without missing values, and computes initial level and total growth.
- Produces a scatter plot of initial level versus total growth with labeled axes.
- States whether the relation is positive or not and labels the five top growth and five highest initial level municipalities.

### Question 1.4

- Correctly finds the pre-2008 peak for each municipality.
- Correctly computes the change from this peak to the latest observation and identifies municipalities still below the peak.
- Provides a simple figure that visualizes these changes and highlights municipalities that have not recovered.

## Problem 2: Exchange Economy

### Question 2.1

- Correctly computes and plots demand for good 1.
- Correctly computes and plots excess demand for good 1.
- Identifies that there are three equilibria.

**Question 2.2**

- Implements the tâtonnement algorithm with the given tolerance, step size and iteration cap.
- Runs it for the initial guesses and plots price and excess demand over iterations.
- States correctly that the initial guess affects which equilibrium is reached.

**Question 2.3**

- Correctly applies the tâtonnement algorithm for the grid of initial guesses.
- Records the converged value (or flags non convergence) and plots  $p_1^0$  vs  $p_1^*$ .
- Comments briefly on which equilibria are found and how the attraction regions look.

**Question 2.4**

- Correctly implements the dampened Newton-Raphson algorithm with numerical derivative, damping and non-negative price constraint.
- Runs it for the grid of initial guesses and plots  $p_1^0$  versus the resulting  $p_1^*$ .
- Briefly compares convergence and equilibria found to the tâtonnement results.

**Problem 3: The AS AD Model****Question 3.1**

- Computes AD and SRAS curves for given parameters and inflation expectations equal to target inflation.
- Produces a static plot with the AD curve, the SRAS curve and the long run equilibrium.
- Shifts the SRAS curve correctly when expectations jump to 0.08 and shows the new equilibrium on the same AD curve.

**Question 3.2**

- Sets up a temporary demand shock with  $v_0 = 0.1$  and inflation expectations updating adaptively over 5 periods.
- Computes the corresponding short-run equilibrium output and inflation in each period.
- Shows in one figure how the relevant curves move over the 5 periods so that the equilibrium path is visible.

**Question 3.3**

- Implements the stochastic demand shock process and simulates the model for a 500 periods with adaptive expectations.
- Produces time series plots for output, inflation and the demand shock and reports basic moments (standard deviations and correlation).
- Repeats the simulation for the alternative persistence parameter and comments briefly on the change in moments.